

EMASAM: a Computationally Efficient Sharpness-Aware Minimization via EMA-Guided Perturbations

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Model Generalization in Deep Learning

Training Dataset



Testing Dataset
(Unseen)



Higher Model Accuracy

Lower Model Accuracy

Generalization Gap

*This work focuses on image classification tasks

Optimization Problem

Given a loss function of the training set:

$$L_S(\mathbf{w}) \triangleq \frac{1}{n} \sum_{i=1}^n l(\mathbf{w}, x_i, y_i)$$

l = loss function measuring prediction error

w = model parameters

x_i = input image

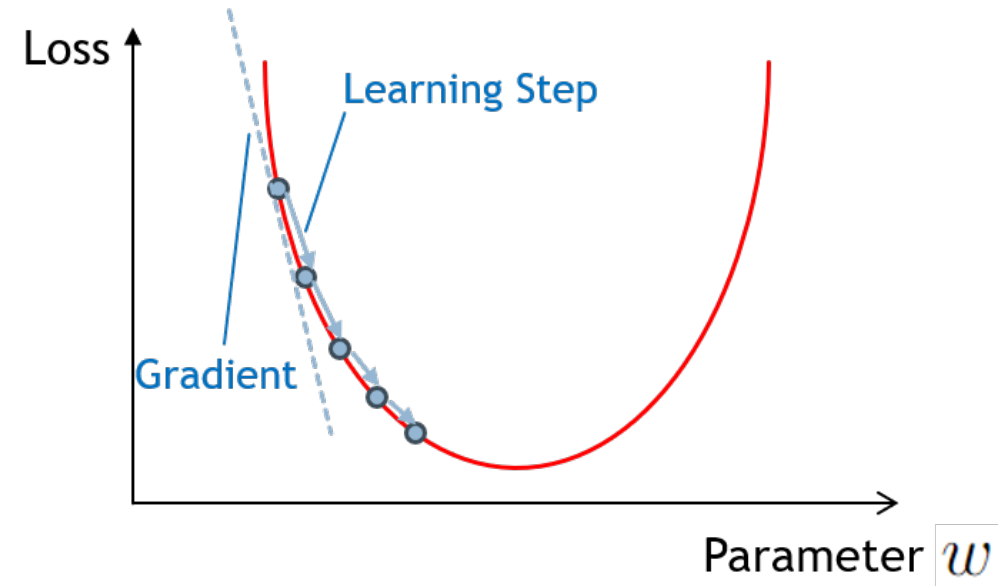
y_i = label

Image classification is considered an optimization problem:

$$\min_{\mathbf{w}} L_S(\mathbf{w})$$

Stochastic Gradient Descent (SGD):

$$w_{t+1}^{\text{SGD}} = w_t - \eta \nabla L_s(w_t) \quad , \eta = \text{learning rate}$$



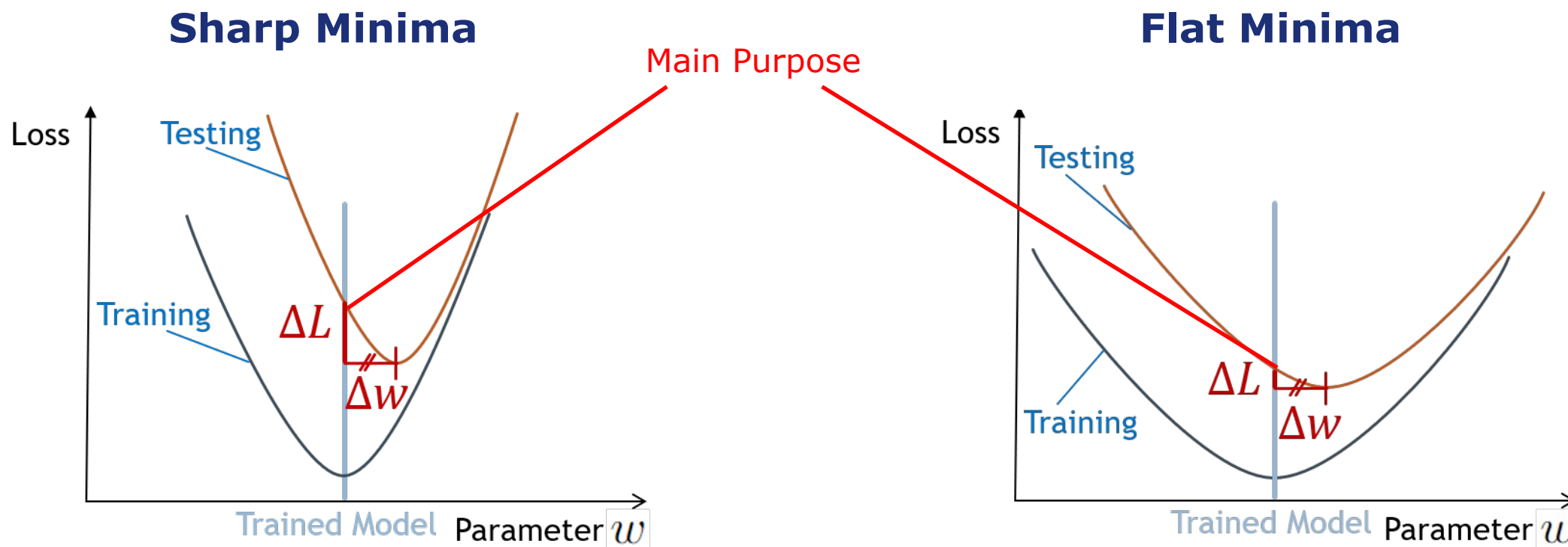
There are also other optimizers,
e.g., RMSprop, Adam, AdamW

Loss Landscape and Generalization

Based on some assumptions:

- The minima of the testing dataset slightly shifts from the training.
- The testing loss landscape has a similar shape to the training.

a sharper loss landscape results in a larger loss increase on the test loss compared to a flat landscape.



Training Methods

Conventional
(SGD)

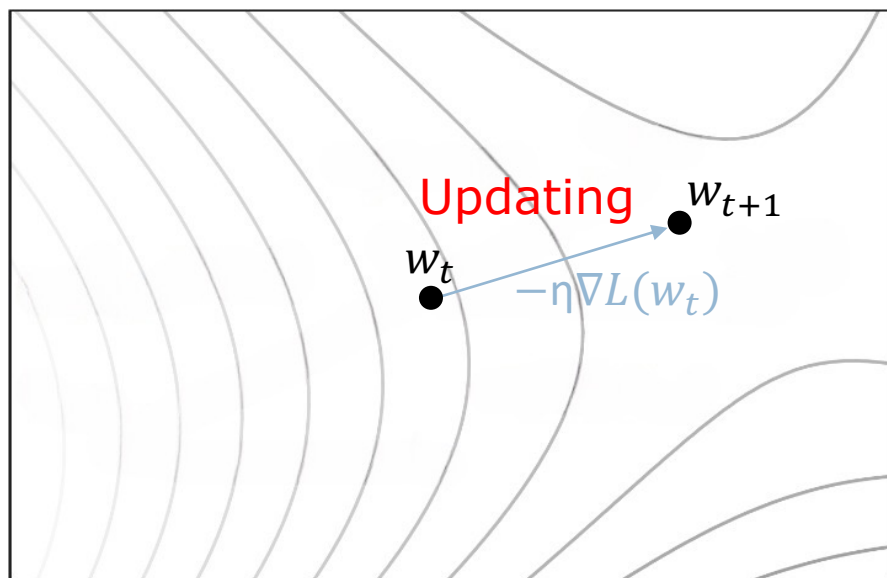
SAM

EMASAM
(Ours)

SAM: Sharpness-Aware Minimization

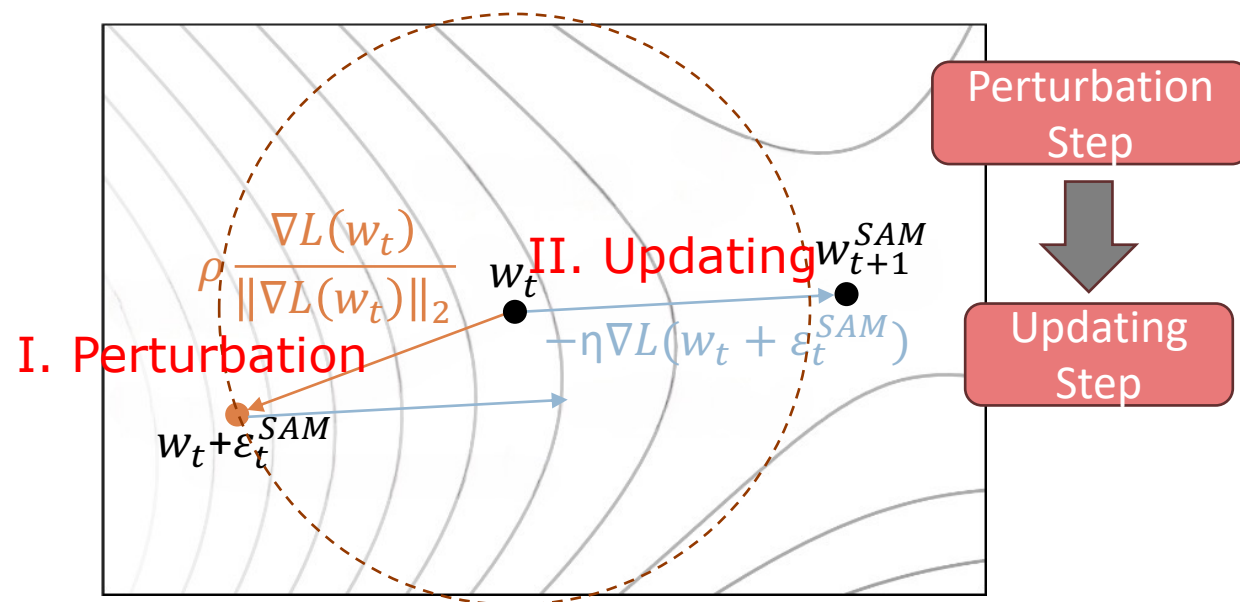
SAM improves generalization by avoiding sharp minima in the loss landscape.

Conventional



$$\min_w L_S(w)$$

SAM

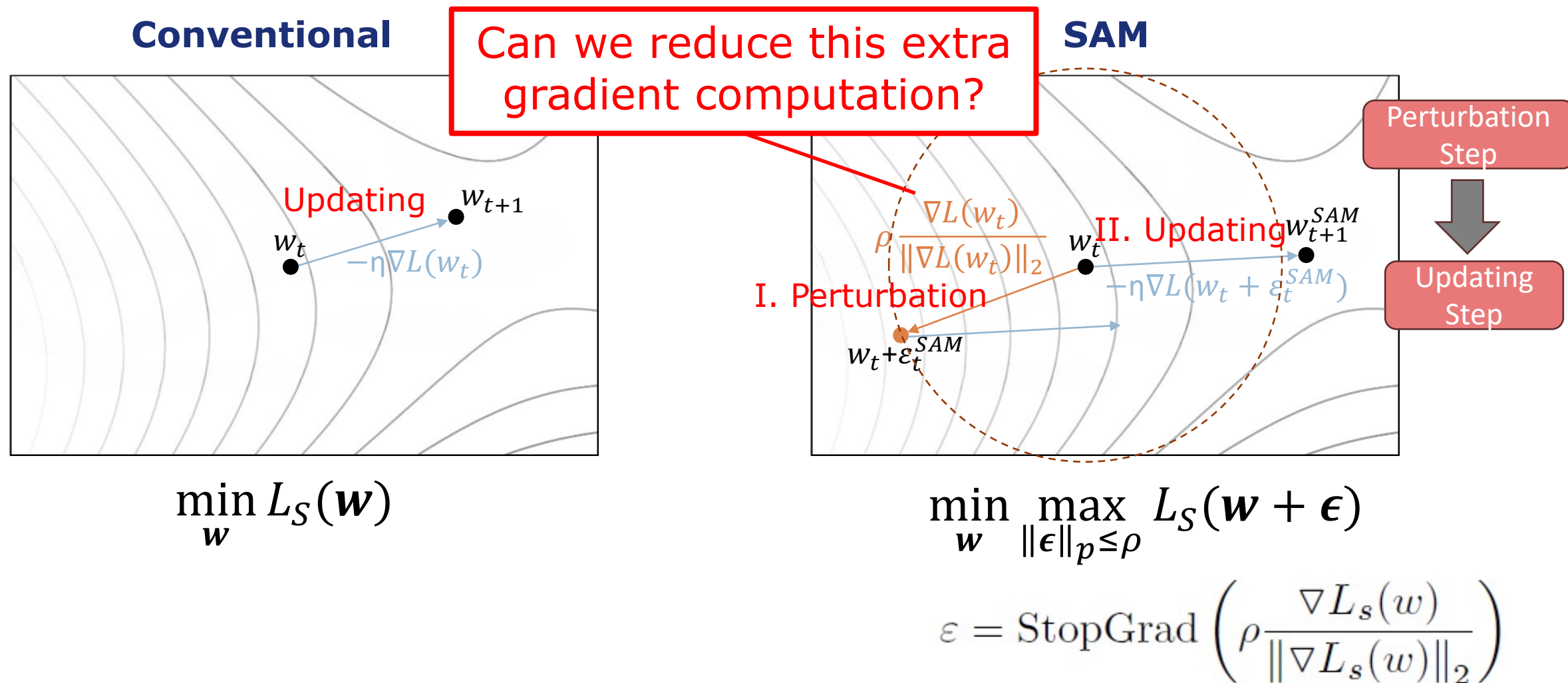


$$\min_w \max_{\|\epsilon\|_p \leq \rho} L_S(w + \epsilon)$$

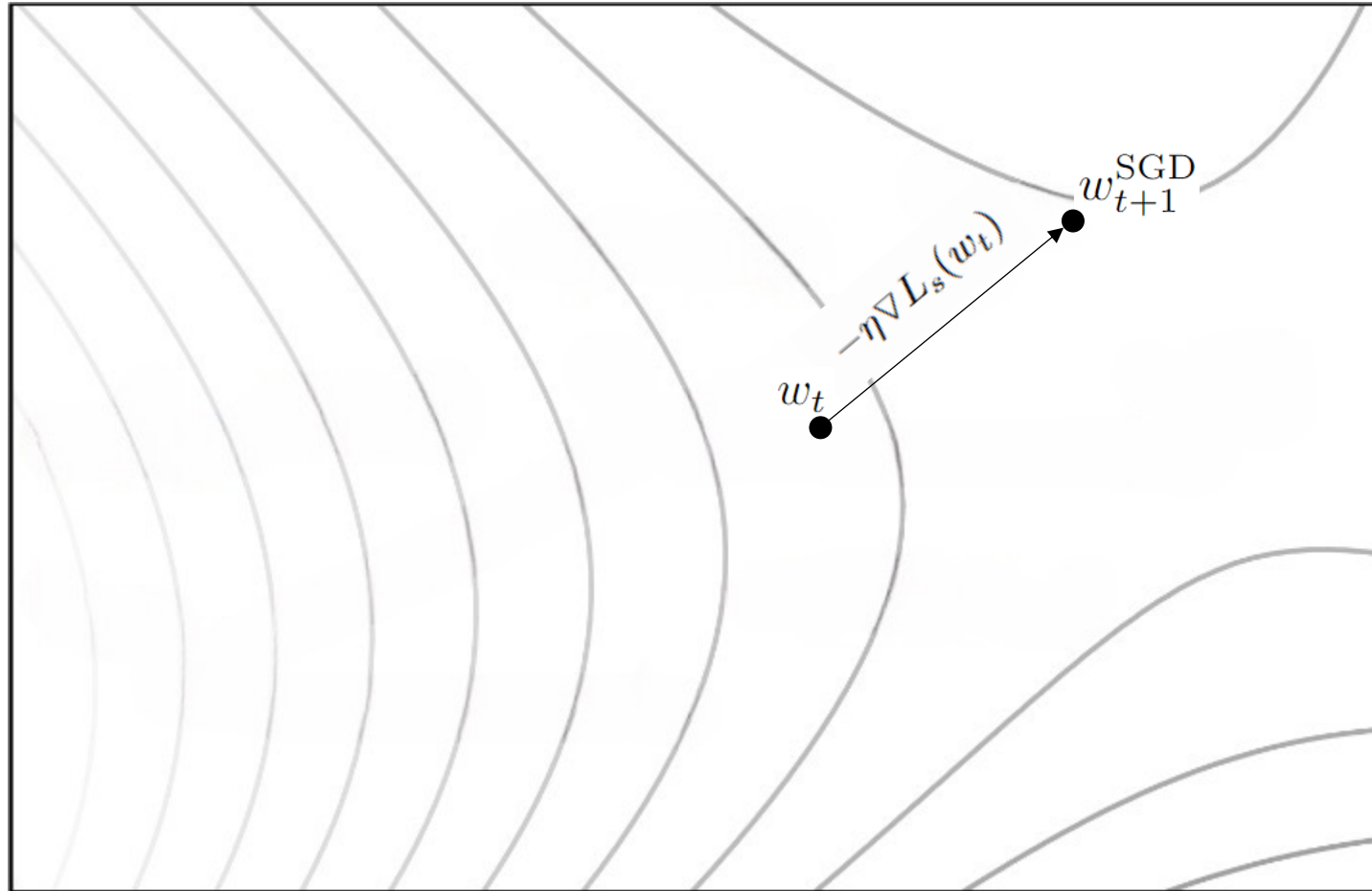
$$\epsilon = \text{StopGrad} \left(\rho \frac{\nabla L_s(w)}{\|\nabla L_s(w)\|_2} \right)$$

SAM: Sharpness-Aware Minimization

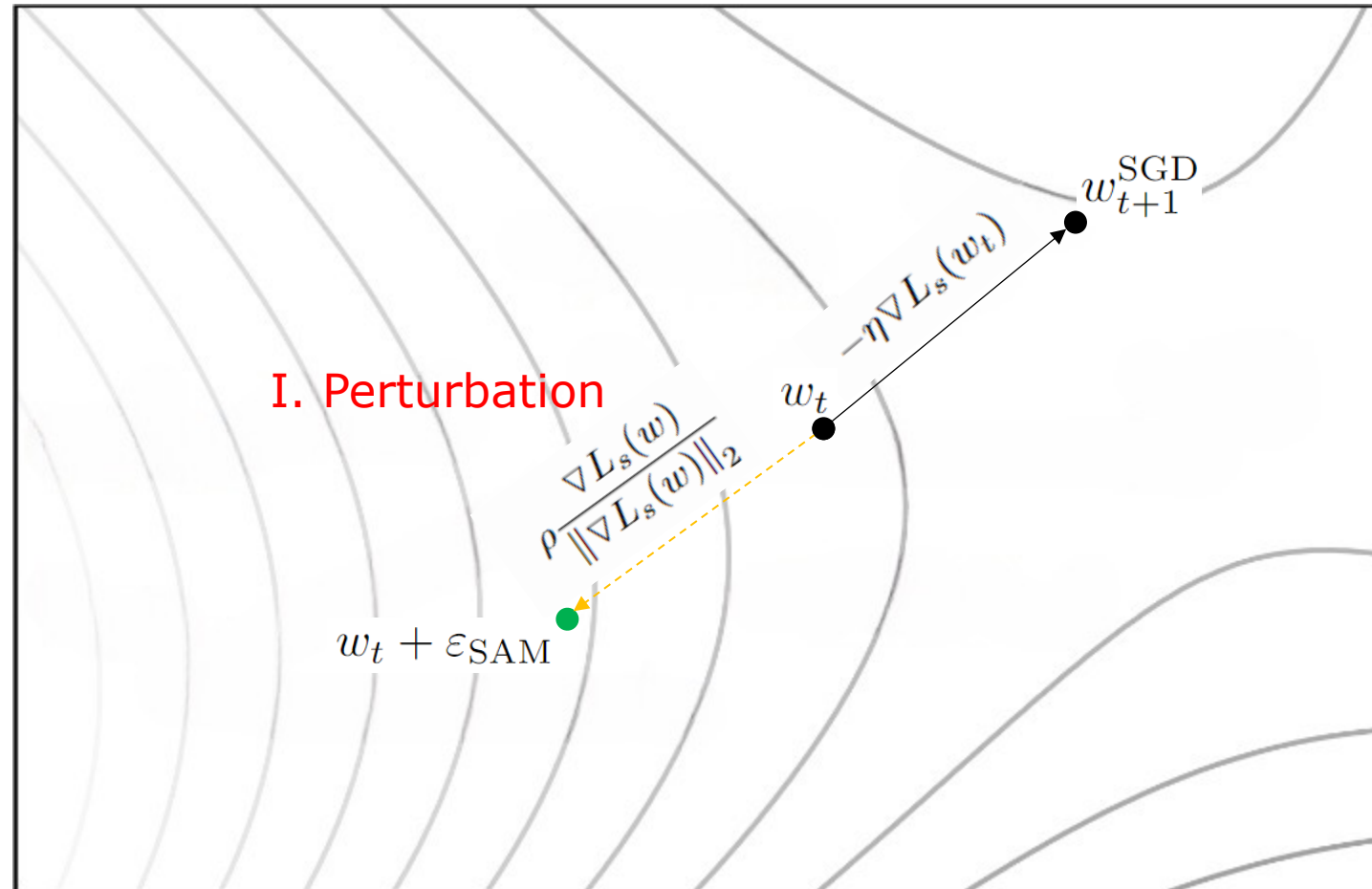
SAM improves generalization by avoiding sharp minima in the loss landscape.



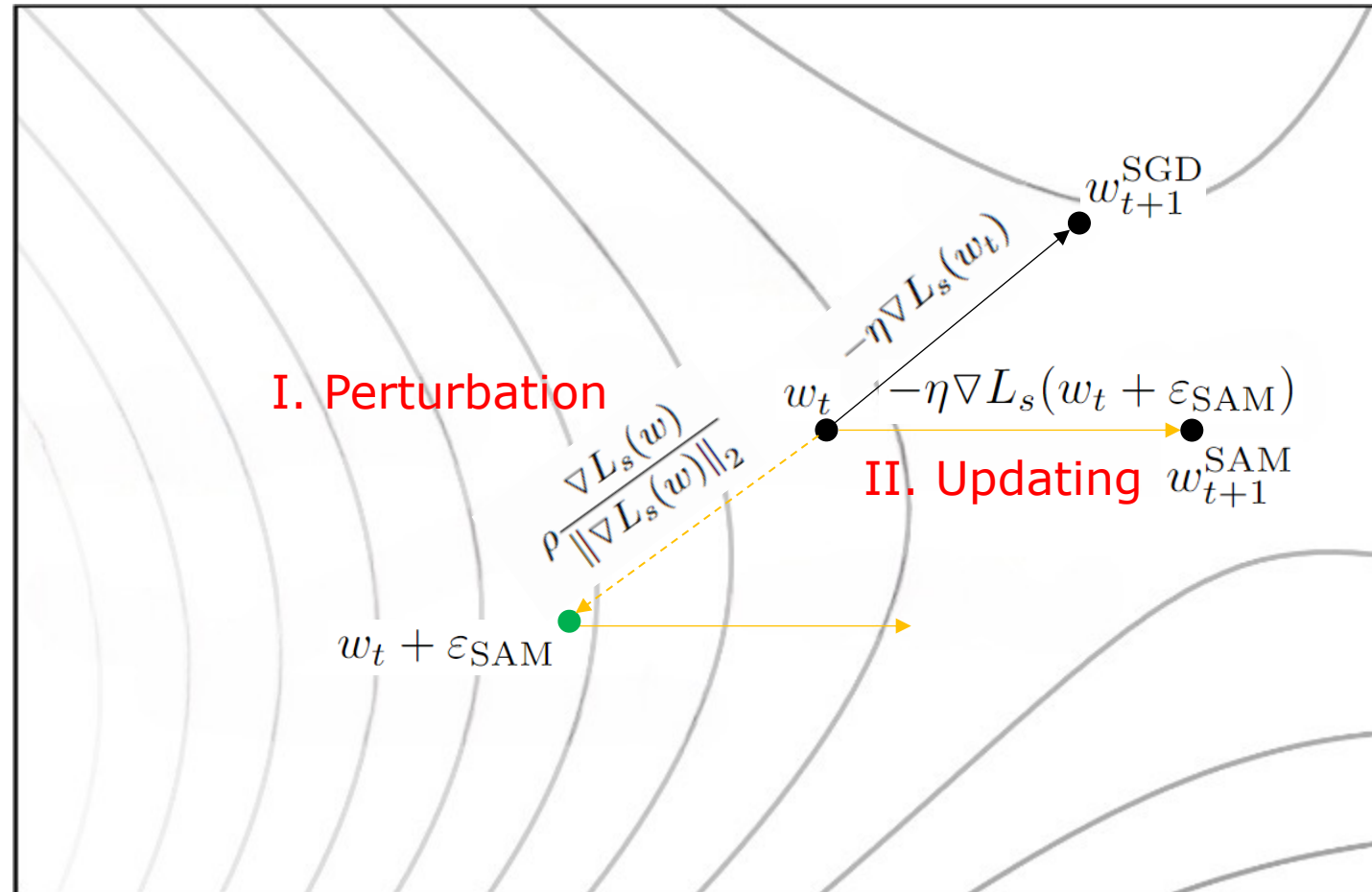
SGD Updating



SAM Updating



SAM Updating



SAM's Two Gradient Calculation

- First Step (Perturbation Step):

$$\varepsilon_{\text{SAM}} = \text{StopGrad} \left(\rho \frac{\nabla L_s(w)}{\|\nabla L_s(w)\|_2} \right)$$

- Second Step (Weight Updating Step):

$$w_{t+1}^{\text{SAM}} = w_t - \eta \nabla L_s(w_t + \varepsilon_{\text{SAM}})$$

Two backpropagations required

Research Question

Can we reduce this extra gradient computation ?

- First Step (Perturbation Step):

$$\varepsilon_{\text{SAM}} = \text{StopGrad} \left(\rho \frac{\nabla L_s(w)}{\|\nabla L_s(w)\|_2} \right)$$

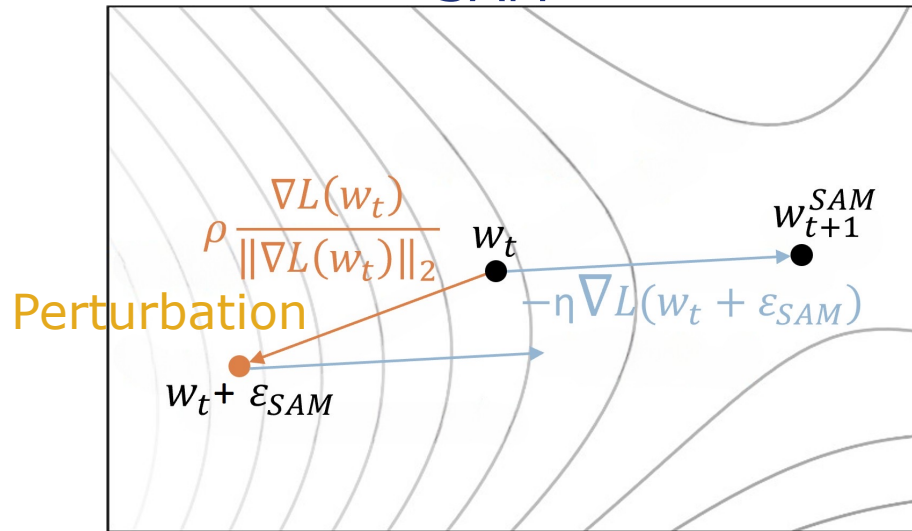
- Second Step (Weight Updating Step):

$$w_{t+1}^{\text{SAM}} = w_t - \eta \nabla L_s(w_t + \varepsilon_{\text{SAM}})$$

Required for weight updating

EXPONENTIAL MOVING AVERAGE SHARPNESS-AWARE MINIMIZATION (EMASAM)

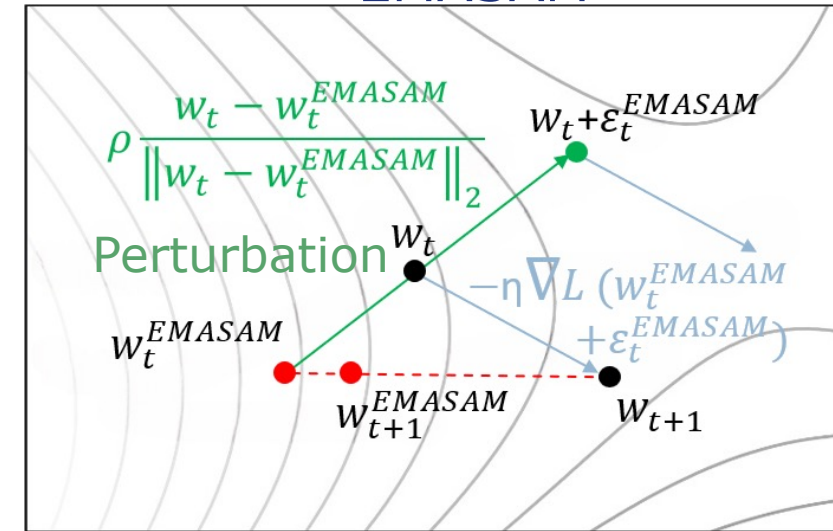
SAM



$$w_{t+1}^{SAM} = w_t - \eta \nabla L_s(w_t + \epsilon_{SAM})$$

- Gradient-based perturbation that requires backpropagation
- Requires **two** backpropagations per training iteration

EMASAM



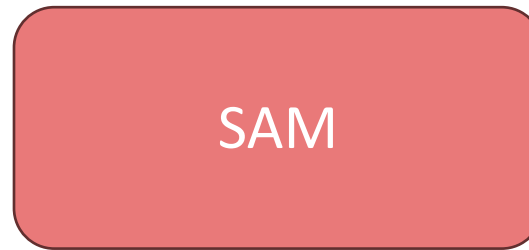
$$w_{t+1}^{EMASAM} = \alpha w_t^{EMASAM} + (1 - \alpha)(w_t - \eta \nabla L(w_t^{EMASAM} + \epsilon_t^{EMASAM}))$$

- EMA-based perturbation that does not require backpropagation
- Requires **one** backpropagation per training iteration

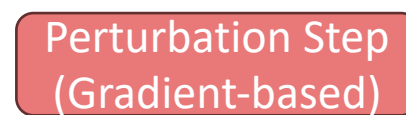
Method Comparison



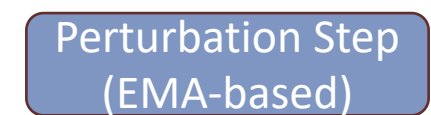
1 Step
1 Gradient Calculation



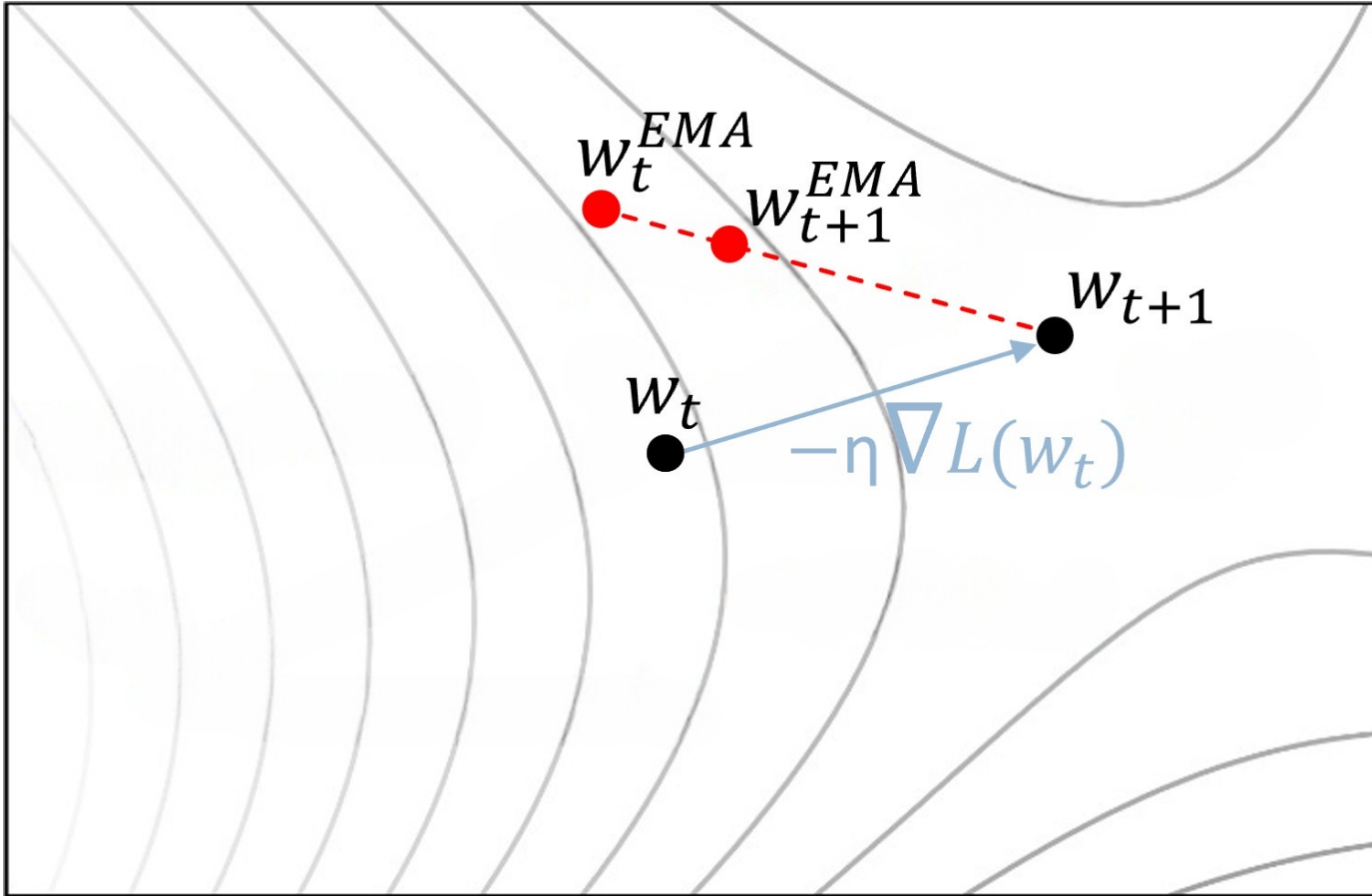
2 Steps
2 Gradient Calculations



2 Steps
1 Gradient Calculations



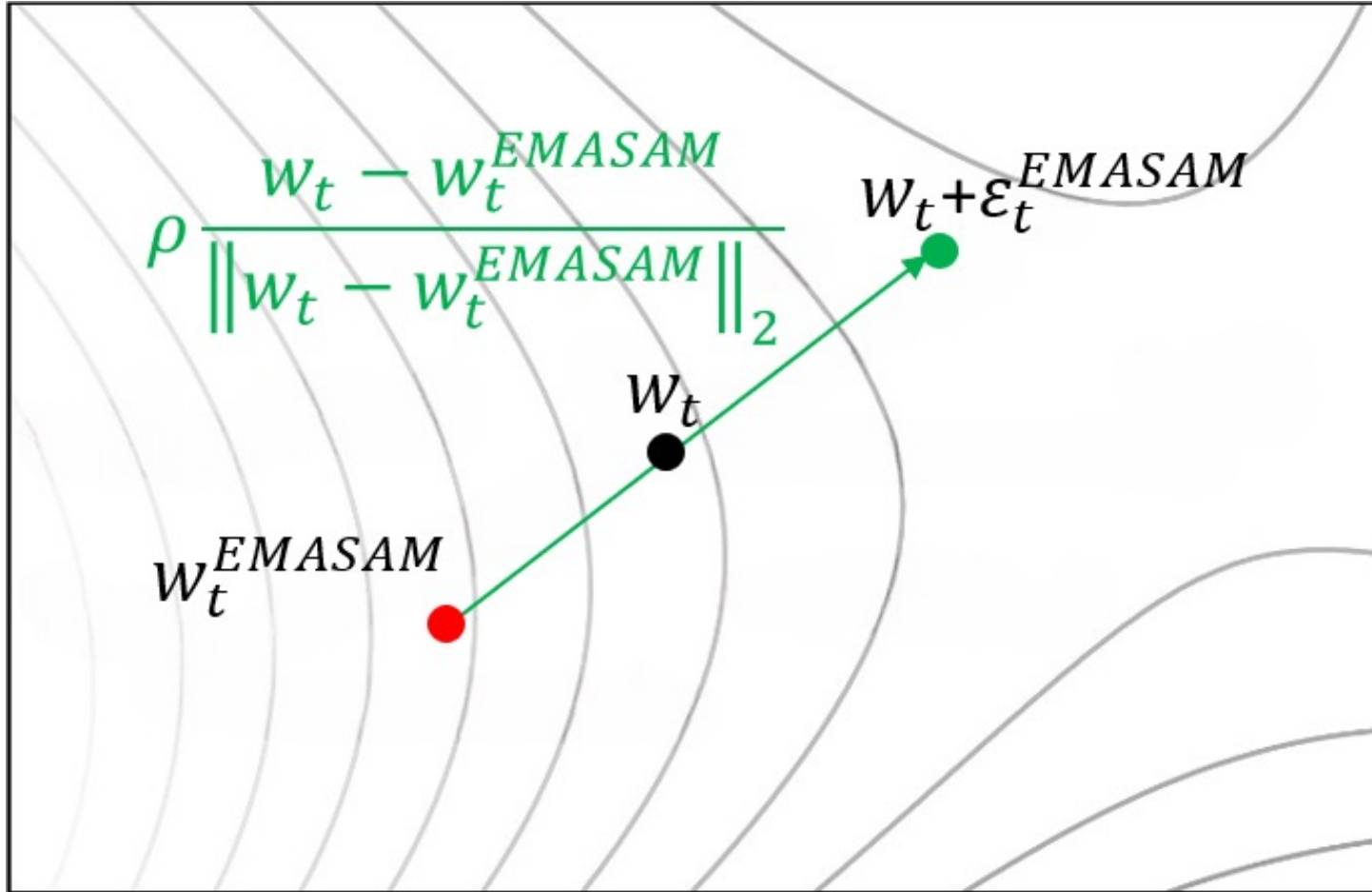
EMA Model



- EMA model is created as a separate copy of the main model.
- The EMA model is updated via an averaging rule at the end of each training iteration

$$w_{t+1}^{EMA} = \alpha w_t^{EMA} + (1 - \alpha) w_{t+1}$$

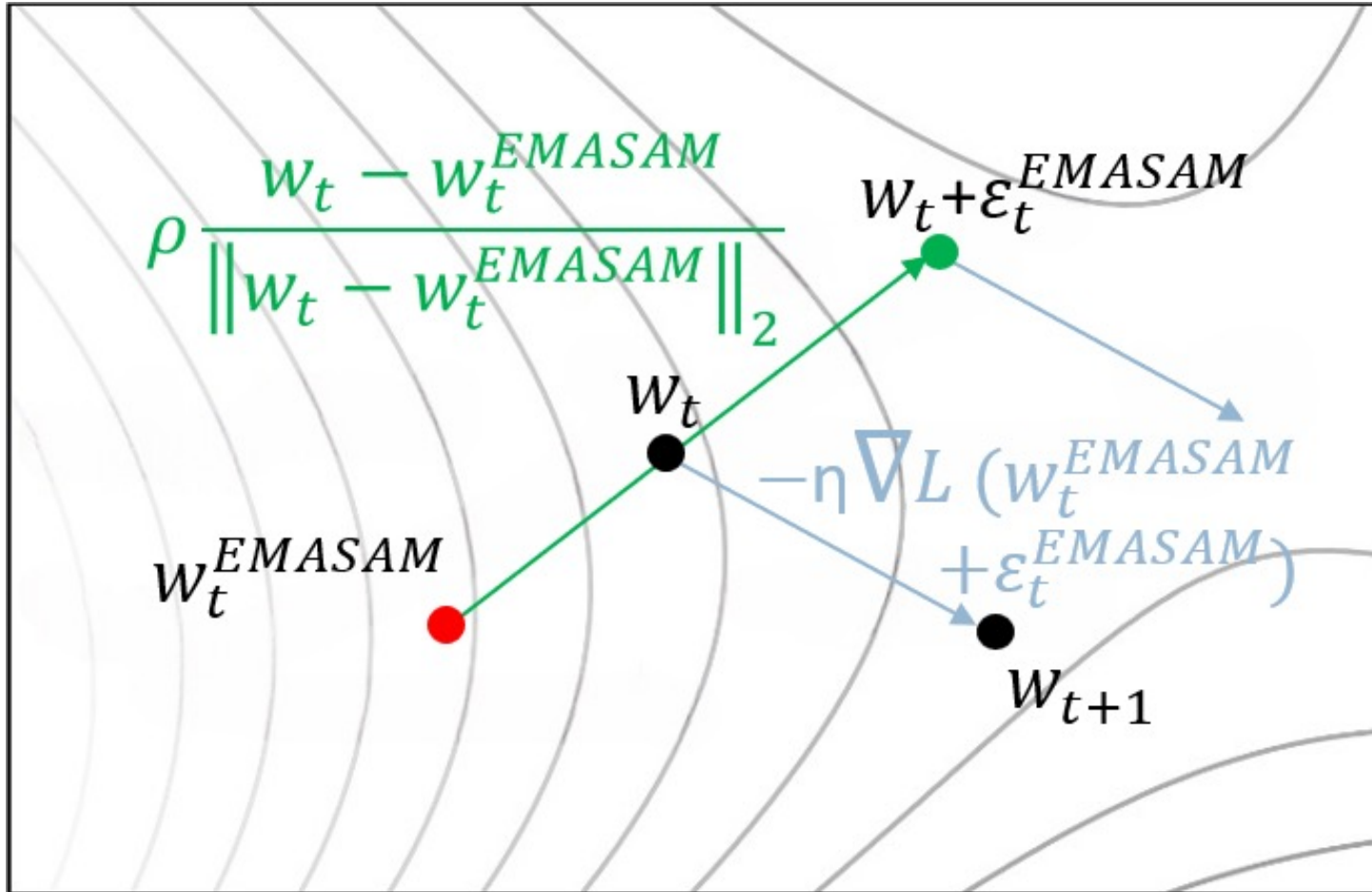
EMASAM Model



- EMASAM perturbs the model using the formula:

$$\epsilon_t^{EMASAM} = \text{StopGrad} \left(\rho \frac{(w_t - w_t^{EMASAM})}{\|w_t - w_t^{EMASAM}\|_2} \right)$$

EMASAM Model

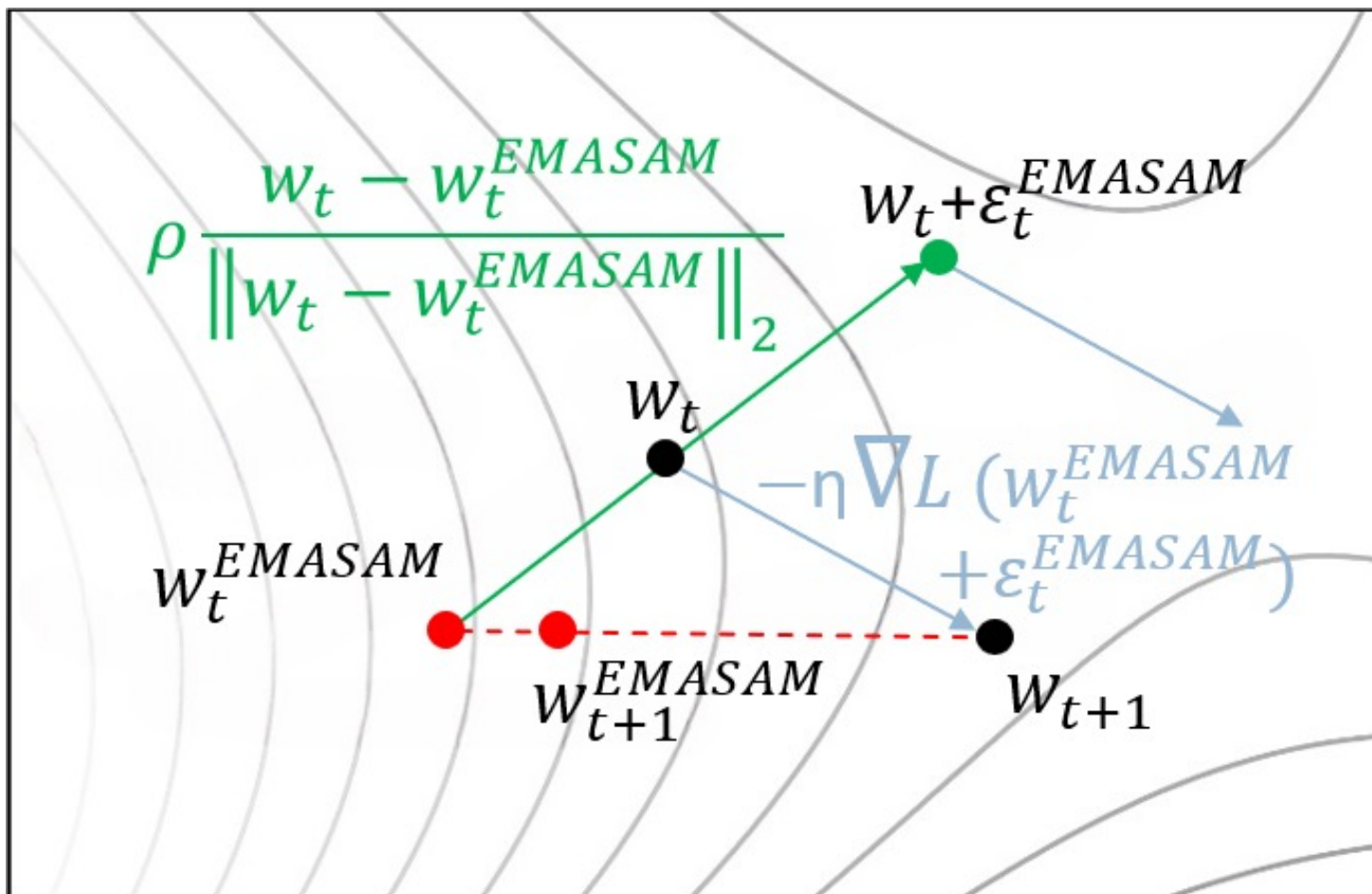


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- At the perturbed position, compute the loss gradient.
- Use the computed gradient to update the original model.

EMASAM Model



- EMASAM perturbs the model using the formula:

$$\epsilon_t^{EMASAM} = \text{StopGrad} \left(\rho \frac{(w_t - w_t^{EMASAM})}{\|w_t - w_t^{EMASAM}\|_2} \right)$$

- At the perturbed position, compute the loss gradient.
- Use the computed gradient to update the original model.
- Update the separate EMA model.

$$\begin{aligned} w_{t+1}^{EMASAM} &= \alpha w_t^{EMASAM} \\ &\quad + (1 - \alpha)(w_t - \eta \nabla L(w_t^{EMASAM} + \epsilon_t^{EMASAM})) \end{aligned}$$

Algorithm 1: EMASAM

Input: training data $\mathcal{S}$, learning rate η , perturbation radius ρ , EMA coefficient α

Initialize weights w_0 , EMA weights w_0^{EMASAM}

for $t = 0$ **to** $T - 1$ **do**

 Sample mini-batch $\mathcal{B}_t \subset \mathcal{S}$

 Perturb weight: $w_t^{\text{EMASAM}} + \varepsilon_t^{\text{EMASAM}} \leftarrow w_t^{\text{EMASAM}} + \rho \frac{(w_t - w_t^{\text{EMASAM}})}{\|w_t - w_t^{\text{EMASAM}}\|_2}$

 Compute gradient: $\nabla L(w_t^{\text{EMASAM}} + \varepsilon_t^{\text{EMASAM}})$

 Update main model: $w_{t+1} \leftarrow w_t - \eta \nabla L(w_t^{\text{EMASAM}} + \varepsilon_t^{\text{EMASAM}})$

 Update EMASAM model: $w_{t+1}^{\text{EMASAM}} \leftarrow \alpha w_t^{\text{EMASAM}} + (1 - \alpha) w_{t+1}$

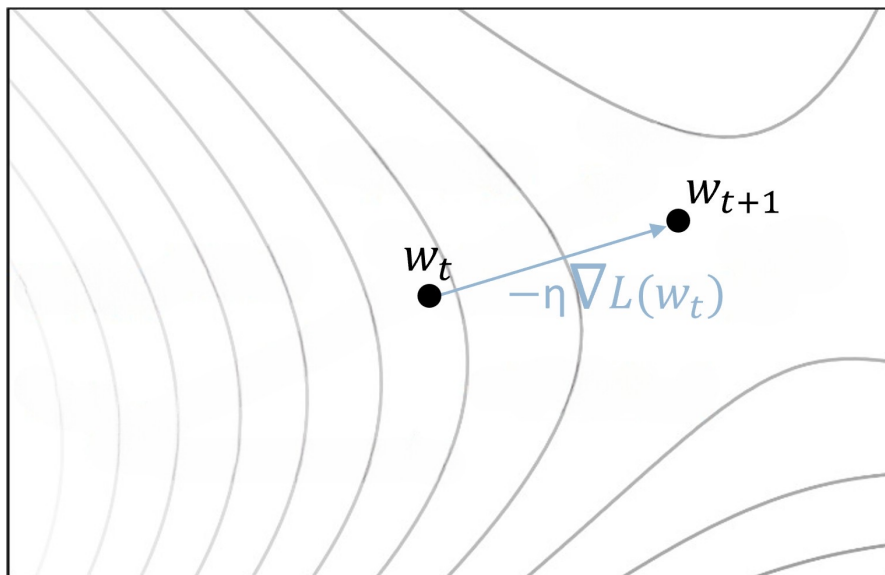
end

return $w_{t+1}, w_{t+1}^{\text{EMASAM}}$

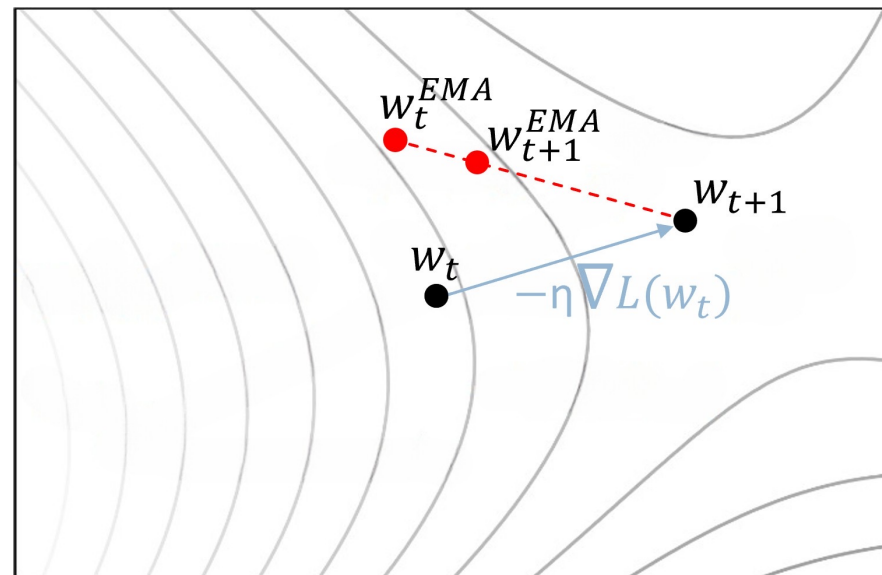
used for evaluation/inference

Method Comparison

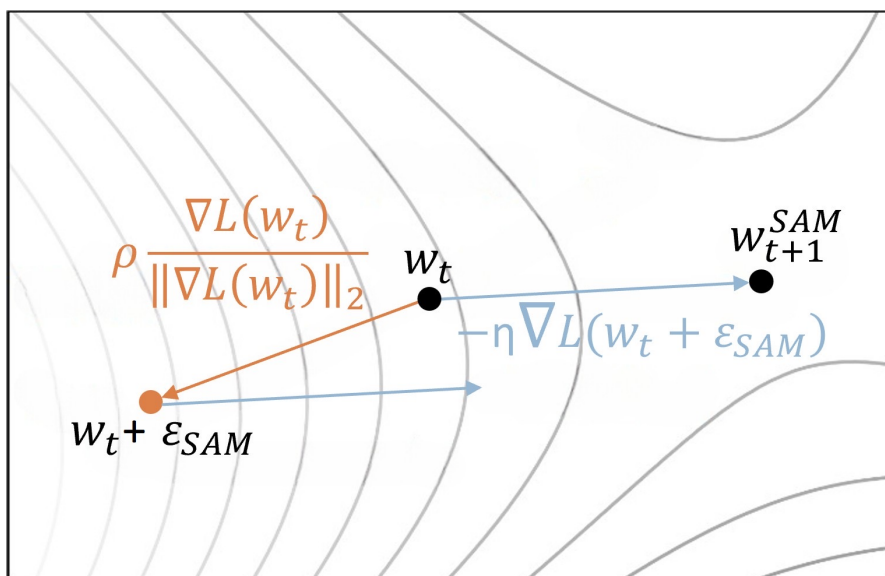
SGD



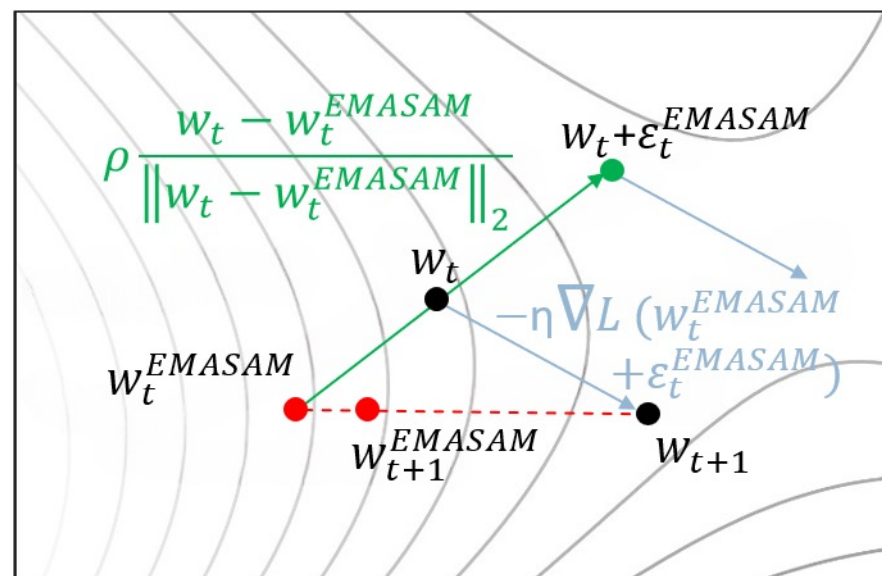
EMA



SAM



EMASAM



Experiments

Hyperparameter tuning

Table 1: Accuracies (%) for hyperparameter tuning of EMASAM on ResNet-50 on CIFAR-100

ρ	with normalization	w/o normalization
0.01	81.74 $\pm$ 0.24	80.85 $\pm$ 0.33
0.02	81.44 $\pm$ 0.13	81.03 $\pm$ 0.39
0.05	81.46 $\pm$ 0.26	80.55 $\pm$ 0.17
0.1	81.26 $\pm$ 0.47	81.38 $\pm$ 0.24
0.2	81.24 $\pm$ 0.17	81.15 $\pm$ 0.15
0.5	81.65 $\pm$ 0.32	80.58 $\pm$ 0.58
1.0	81.74 $\pm$ 0.37	81.46 $\pm$ 0.18
2.0	82.34$\pm$0.54	81.71 $\pm$ 0.37
5.0	<u>81.82$\pm$0.17</u>	81.54 $\pm$ 0.24

Bold = The best accuracy, Underline = Second best accuracy

Experiments

Accuracy Comparisons using ResNet-18.

Table 2: Accuracies (%) of models trained with ResNet-18

Method	CIFAR-10	CIFAR-100	Fashion-MNIST	EMNIST	Average
Vanilla (SGD)	94.57±0.07	77.62±0.47	89.81±0.78	88.07±0.24	87.52
SAM [11]	<u>96.09±0.03</u>	<u>79.80±0.29</u>	92.21±0.14	90.11±0.02	<u>89.55</u>
SS-SAM [39]	95.64±0.31	79.34±0.85	91.93±0.74	89.75±0.17	89.17
AE-SAM	95.79±0.08	78.71±0.55	<u>92.39±0.19</u>	90.02±0.12	89.23
MSAM [1]	94.84±0.14	78.18±0.16	92.21±0.21	89.39±0.48	88.65
EMA	96.04±0.11	79.25±0.25	90.98±0.14	88.01±0.15	88.57
EMA + RandPerturb	93.34±0.03	74.79±0.13	91.60±0.16	89.93±0.11	87.42
EMASAM [Ours]	96.23±0.09	79.87±0.17	93.36±0.11	<u>90.08±0.10</u>	89.88

Bold = The best accuracy, Underline = Second best accuracy

* Each setup was done 3 times, the mean accuracy and standard deviation are reported.

Experiments

Accuracy Comparisons using ResNet-50.

Table 3: Accuracies (%) of models trained with ResNet-50

Method	CIFAR-10	CIFAR-100	Fashion-MNIST	EMNIST	Average
Vanilla (SGD)	94.56 $\pm$ 0.15	77.21 $\pm$ 0.09	90.87 $\pm$ 1.03	88.44 $\pm$ 0.16	87.77
SAM [11]	96.00 $\pm$ 0.13	79.41 $\pm$ 0.61	<u>92.33$\pm$0.24</u>	89.25 $\pm$ 0.60	89.25
SS-SAM [39]	95.63 $\pm$ 0.18	79.91 $\pm$ 0.26	92.22 $\pm$ 0.00	89.43 $\pm$ 0.11	<u>89.30</u>
AE-SAM [16]	95.45 $\pm$ 0.33	78.82 $\pm$ 1.32	92.18 $\pm$ 0.09	89.02 $\pm$ 0.11	88.87
MSAM [1]	93.16 $\pm$ 0.21	78.16 $\pm$ 0.73	90.41 $\pm$ 0.24	88.41 $\pm$ 0.53	87.54
EMA	<u>96.49$\pm$0.06</u>	<u>81.50$\pm$0.23</u>	90.38 $\pm$ 0.10	87.96 $\pm$ 0.09	89.08
EMA + RandPerturb	92.92 $\pm$ 0.03	74.90 $\pm$ 0.46	91.20 $\pm$ 0.07	<u>89.61$\pm$0.07</u>	87.16
EMASAM [Ours]	96.78$\pm$0.20	82.34$\pm$0.54	92.57$\pm$0.10	89.91$\pm$0.01	90.40

Bold = The best accuracy, Underline = Second best accuracy

* Each setup was done 3 times, the mean accuracy and standard deviation are reported.

Experiments

Accuracy Comparisons using WideResNet-28-10.

Table 4: Accuracies (%) of models trained with WideResNet-28-10

Method	CIFAR-10	CIFAR-100	Fashion-MNIST	EMNIST	Average
Vanilla (SGD)	95.83±0.11	81.50±0.27	89.54±0.88	86.67±0.38	88.39
SAM [11]	<u>96.96±0.05</u>	84.05±0.27	91.63±0.26	<u>89.58±0.13</u>	<u>90.55</u>
SS-SAM [39]	96.75±0.15	<u>83.03±0.09</u>	91.54±0.26	89.74±0.15	90.26
AE-SAM [16]	<u>96.96±0.08</u>	83.40±0.10	<u>91.89±0.40</u>	89.55±0.13	90.45
MSAM [1]	95.74±0.02	82.27±0.13	91.63±0.40	87.89±0.32	89.38
EMA	96.70±0.01	81.96±0.02	90.39±0.23	86.10±0.19	88.79
EMA + RandPerturb	94.73±0.06	77.75±0.17	91.76±0.03	90.03±0.11	88.57
EMASAM [Ours]	97.13±0.13	82.96±0.04	92.94±0.09	89.42±0.02	90.61

Bold = The best accuracy, Underline = Second best accuracy

* Each setup was done 3 times, the mean accuracy and standard deviation are reported.

Experiments

Accuracy Comparisons using PyramidNet-110.

Table 5: Accuracies (%) of models trained with PyramidNet-110

Method	CIFAR-10	CIFAR-100	Fashion-MNIST	EMNIST	Average
Vanilla (SGD)	94.08±0.04	75.92±0.39	87.84±1.00	89.03±0.20	86.72
SAM [11]	95.91±0.04	79.15±0.52	<u>91.53±0.30</u>	89.16±0.27	88.92
SS-SAM [39]	95.49±0.25	80.19±0.16	91.02±0.96	88.41±0.28	88.58
AE-SAM [16]	96.10±0.46	79.22±0.90	91.47±0.33	89.38±0.10	<u>89.04</u>
MSAM [1]	93.61±0.35	77.49±0.25	90.32±0.30	88.39±0.26	87.45
EMA	<u>96.58±0.05</u>	<u>80.74±0.41</u>	89.48±0.38	86.78±0.21	88.40
EMA + RandPerturb	93.63±0.11	73.19±0.19	91.02±0.01	89.86±0.07	86.22
EMASAM [Ours]	96.77±0.05	81.86±0.27	92.56±0.10	<u>89.69±0.06</u>	90.22

Bold = The best accuracy, Underline = Second best accuracy

* Each setup was done 3 times, the mean accuracy and standard deviation are reported.

Experiments

Accuracy Comparisons using ResNet-50 on ImageNet-1k dataset.

Table 6: Accuracies (%) of models trained with ResNet-50 on ImageNet-1k

Method	Top-1	Top-5
Vanilla (SGD)	76.45	93.22
SAM [11]	<u>76.71</u>	93.32
SS-SAM [39]	76.64	93.48
AE-SAM [16]	76.68	<u>93.41</u>
MSAM [1]	75.83	92.87
EMA	76.51	92.87
EMA + RandPerturb	60.58	83.07
EMASAM [Ours]	76.87	93.23

Bold = The best accuracy, Underline = Second best accuracy

Experiments

Accuracy Comparisons using ViT-B/16 on ImageNet-1k dataset.

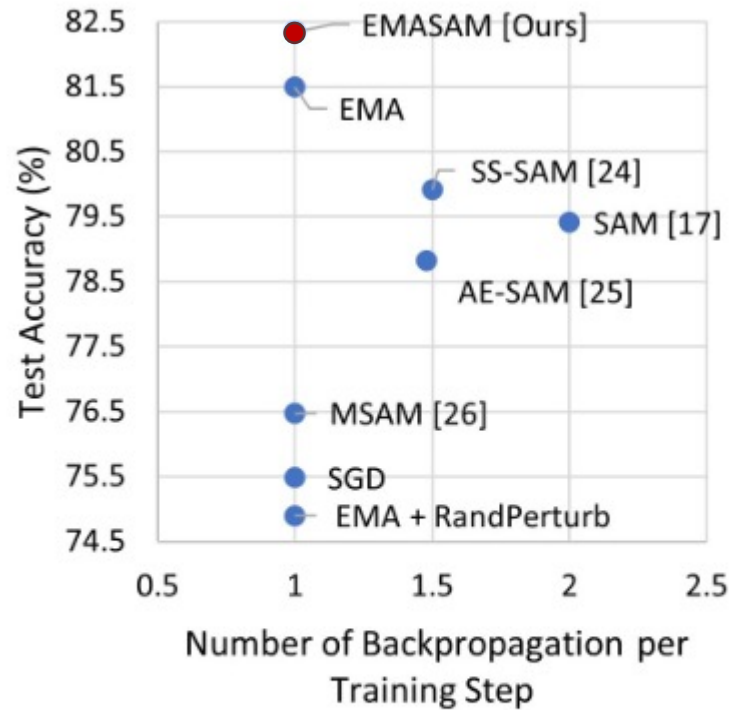
Table 7: Accuracies (%) of models trained with ViT-B/16 on ImageNet-1k

<i>Method</i>	Top-1	Top-5
Vanilla (AdamW)	69.16	87.79
SAM [11]	71.19	89.21
SS-SAM [39]	69.58	88.28
AE-SAM [16]	72.06	<u>89.97</u>
MSAM [1]	71.59	89.48
EMA	71.04	89.42
EMA + RandPerturb	46.67	70.12
EMASAM [Ours]	<u>72.00</u>	90.15

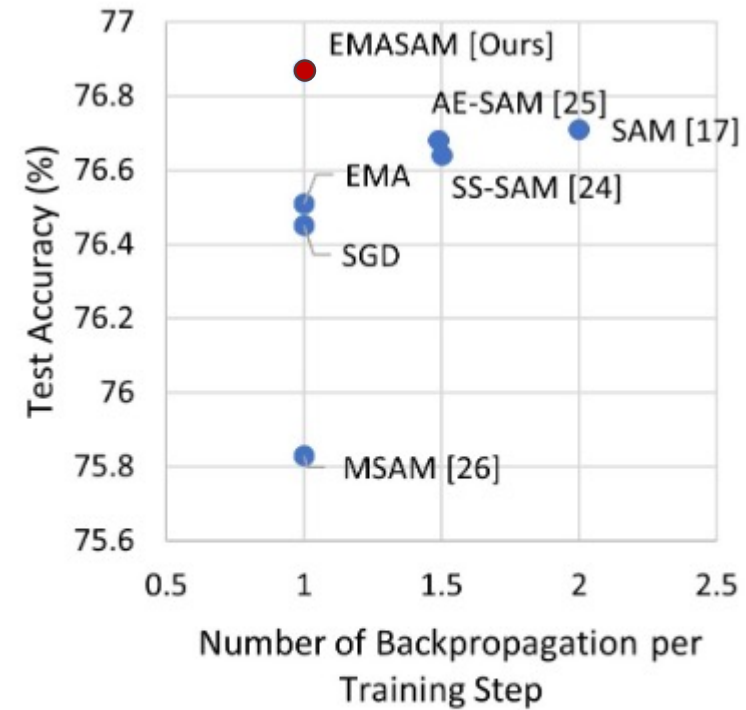
Bold = The best accuracy, Underline = Second best accuracy

Experiments

Number of backpropagation VS test accuracy



(a) ResNet-50 on CIFAR-100



(b) ResNet-50 on ImageNet-1k

Number of backpropagation and test accuracy comparison among models trained with ResNet-50 on CIFAR-100 and ImageNet-1k.
(upper-left → more favorable)

- This research proposed **EMASAM**, which replaces the gradient-based perturbation with EMA-guided perturbation
- EMASAM removes extra gradient calculation in the perturbation step.
- Empirical results prove that EMASAM helps reduce the computational cost while keeping the generalizability of the SAM-like algorithm.

Future work

- Evaluation on more datasets and models.
- Extending to other applications such as object detection or segmentation.
- Investigation on other perturbation strategies that don't require an additional hyperparameter.

[Reproduction Code
and Project Details](#)



Thank You

